

1 Lagrange Interpolation

Let F be a field and let $x_1, \dots, x_n \in F$ be distinct. Given values $y_1, \dots, y_n \in F$, the interpolation problem is to find the unique polynomial $p(x) \in F[x]$ of degree at most $n - 1$ satisfying $p(x_i) = y_i$ for all i .

For each i , define the Lagrange basis polynomial

$$L_i(x) = \prod_{\substack{j=1 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}.$$

Since the x_i are distinct, all denominators are nonzero and hence invertible in F . These basis polynomials satisfy

$$L_i(x_j) = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Consequently, the unique interpolating polynomial is

$$p(x) = \sum_{i=1}^n p(x_i) L_i(x) = \sum_{i=1}^n y_i L_i(x).$$

Indeed, $p(x_k) = y_k$ for every k . Uniqueness follows from the fact that a nonzero polynomial of degree at most $n - 1$ cannot have n distinct roots.

More generally, the value of p at any $a \in F$ can be expressed directly in terms of its values at the interpolation points:

$$p(a) = \sum_{i=1}^n y_i L_i(a).$$

In particular, setting $a = 0$ gives

$$p(0) = \sum_{i=1}^n \lambda_i y_i, \quad \lambda_i = L_i(0) = \prod_{\substack{j=1 \\ j \neq i}}^n \frac{-x_j}{x_i - x_j}.$$

The coefficients λ_i depend only on the public interpolation points and not on the polynomial values y_i .

2 Multiplication over Shamir Shared Secrets

Suppose s_1 and s_2 are shared using degree- $(t - 1)$ polynomials $f(x)$ and $g(x)$, respectively, with $f(0) = s_1$ and $g(0) = s_2$. Party P_i , associated with the public point x_i , holds $f(x_i)$ and $g(x_i)$.

$$h(x_i) = f(x_i)g(x_i).$$

Thus, the resulting values are evaluations of

$$h(x) = f(x)g(x).$$

Since $\deg(f), \deg(g) \leq t - 1$, we have

$$\deg(h) \leq 2t - 2.$$

Moreover,

$$h(0) = f(0)g(0) = s_1s_2.$$

The product is therefore already encoded in $h(x)$, but the degree is too high. We want to construct a new polynomial of degree at most $t - 1$ that has the same secret, namely the same value at $x = 0$.

2.1 Degree Reduction

Let P_j hold $h(x_j)$. Party P_j chooses a fresh random polynomial $q_j(x)$ of degree at most $t - 1$ such that

$$q_j(0) = h(x_j).$$

It distributes $q_j(x_i)$ to party P_i . The value $q_j(0)$ is therefore not revealed; it is merely shared among the parties.

Since $h(x)$ has degree at most $2t - 2$ and we assume $2t - 2 \leq n$, its value at zero can be expressed using the n public points $x_1, \dots, x_n$ through Lagrange interpolation:

$$h(0) = \sum_{j=1}^n L_j(0)h(x_j).$$

The coefficients $L_j(0)$ depend only on the public points and can therefore be computed by every party.

We now define the new sharing polynomial

$$r(x) = \sum_{j=1}^n L_j(0)q_j(x),$$

where $q_j(x)$ is the fresh degree- $(t - 1)$ polynomial chosen by P_j such that $q_j(0) = h(x_j)$.

Each party P_i obtains its new share by evaluating $r(x)$ at its public point x_i :

$$r(x_i) = \sum_{j=1}^n L_j(0)q_j(x_i).$$

2.2 Correctness

The construction satisfies three essential properties.

1. The degree is reduced. Each $q_j(x)$ has degree at most $t - 1$. Therefore, their linear combination also has degree at most $t - 1$:

$$\deg(r) \leq t - 1.$$

Thus, the resulting values form a valid degree- $(t - 1)$ Shamir sharing.

2. The secret is preserved. Evaluating $r(x)$ at zero gives, $r(0) = \sum_{j=1}^n L_j(0)q_j(0)$.

Since $q_j(0) = h(x_j)$, $r(0) = \sum_{j=1}^n L_j(0)h(x_j)$. So, by Lagrange interpolation,

$$r(0) = h(0) = s_1s_2.$$

Hence, $r(x)$ is a degree- $(t - 1)$ sharing polynomial for the product.

3. The secret is not reconstructed. Although $r(0)$ is mathematically equal to s_1s_2 , the parties do not compute it explicitly. Each party only holds $r(x_i)$. To obtain $r(0)$, t parties would have to come together and interpolate $r(x)$.

Thus, multiplication transforms the high-degree sharing $h(x)$ into a fresh degree- $(t - 1)$ sharing $r(x)$ of the same secret, without revealing the product during the protocol.

2.3 Interpretation

The role of Lagrange interpolation is particularly important here. We do not use it to reconstruct $h(0)$ publicly. Instead, the coefficients $L_j(0)$ tell us how the values $h(x_j)$ contribute to the secret. Each such value is first re-shared through $q_j(x)$, and the same Lagrange coefficients are then used to combine these re-sharings.

Consequently, multiplication consists of two steps: local multiplication increases the degree, while Lagrange-based re-sharing reduces the degree back to $t - 1$ while preserving the secret.

3 BGW Protocol

The multiplication procedure over Shamir shares, together with local addition, enables the parties to evaluate arbitrary arithmetic circuits without revealing their private inputs. This is the basic idea of the BGW protocol, named after Ben-Or, Goldwasser, and Wigderson [1]. BGW provides information-theoretic security in the honest-majority setting by maintaining Shamir sharings throughout the computation.

3.1 Arithmetic Circuits

Let $f(x, y) \in \mathbb{F}[x, y]$. An arithmetic circuit for f is a directed acyclic graph with input wires x, y , constant wires, and gates labeled by addition or multiplication. Each gate computes

$$z = u + v \quad \text{or} \quad z = uv.$$

The output wire evaluates to $f(x, y)$.

For example, for $f(x, y) = (x + y)(x + 2y)$, the circuit computes $x + y$ and $x + 2y$ using addition gates and then multiplies the two results.

3.2 Idea Overview of BGW

BGW maintains the invariant that every wire is represented by a degree- $(t - 1)$ Shamir sharing polynomial. The two types of gates are handled as follows:

1. Addition $\longrightarrow$ local addition
2. Multiplication $\longrightarrow$ local multiplication + degree reduction

Addition preserves the polynomial degree, whereas multiplication increases the degree to at most $2t - 2$. The degree-reduction procedure described above restores the sharing to degree $t - 1$ without revealing the intermediate product. Thus, the circuit can be evaluated gate by gate while preserving the same Shamir-sharing structure throughout.

3.3 The Protocol

Given an arithmetic circuit C with private inputs $s_1, \dots, s_m$, the BGW protocol evaluates C directly over Shamir shares. Each input is first shared using a degree- $(t - 1)$ polynomial. The circuit is then evaluated gate by gate: addition gates are computed locally by adding the corresponding shares, while multiplication gates are computed locally followed by the degree-reduction procedure described above. Thus, after every gate, the resulting wire is represented by a degree- $(t - 1)$ Shamir sharing. After evaluating the final gate, the output can be reconstructed by any t parties using Lagrange interpolation, or retained as a sharing if the output is to remain private.

References

- [1] M. Ben-Or, S. Goldwasser, and A. Wigderson, “Completeness Theorems for Non-Cryptographic Fault-Tolerant Distributed Computation,” in *Proceedings of the 20th Annual ACM Symposium on Theory of Computing (STOC)*, pp. 1–10, 1988.